Review 2 Commitment

1. Project Information

Project code	FA26SE032
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Review 2 date	29/10/2026
Reviewer 1	longt5
Reviewer 2	ThinhDP2
Version	2.0
Date	29/10/2026

2. Final Registration Form

Final version	Version 2.0 (Post-Review 2 Scope Realignment)
Date confirmed	29/10/2026
Record of Changes (BM03)	Form BM03 attached with 6 accepted architectural changes.

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-01	User Authentication & RBAC Guard	Customer, Collaborator, Shop Manager, System Manager, System Administrator	Argon2id hashing, JWT Access/Refresh tokens, strict RBAC guard for 5 roles.	WF-01	Nguyễn Thành Thắng
F-02	Identity Verification & KYC Multi-Tier Approval	Collaborator, Shop Manager, System Manager	CCCD / Business license upload, multi-status verification (PENDING/APPROVED/REJECTED) by System Manager.	WF-01	Nguyễn Thành Thắng
F-03	Collaborator Social Portfolio Linking	Collaborator	Link TikTok, Facebook, YouTube channels with follower counts and content category tags.	WF-01	Nguyễn Thành Thắng
F-04	Digital Media Asset Hub	Shop Manager, Collaborator	Centralized promotional asset library with tags, resolution filtering, and one-click download.	WF-01	Nguyễn Thành Thắng
F-05	Real-time Security Audit Logging	System Administrator, System Manager	Immutable recording of sensitive financial, KYC, payout, and dispute actions with actor ID & IP.	WF-01, WF-05	Nguyễn Thành Thắng
F-06	Additive Multi-Rule Commission Engine	Shop Manager, Collaborator, System Manager	Calculates min(25%, Base(10%) + Tier(1-3%) + max(Deal, Campaign) + Live(2-3%)) + MonthlyBonus.	WF-02, WF-03, WF-05	Phan Xuân Thịnh & Nguyễn Đình Tuấn
F-07	Smart Shortened Link & QR Generator	Collaborator	Generates /r/:code links and SVG dynamic QR codes with Redis rate limit protection.	WF-03	Nguyễn Đình Tuấn
F-08	Last-Click Cookie Attribution & Tracking	Customer, Collaborator	30-day cookie window, first/last touch attribution, referrer preservation across sessions.	WF-03	Nguyễn Đình Tuấn
F-09	3-Tier Multi-Store Voucher Stacking	Customer, Shop Manager	Stack Global + Store (couponsByStore) + Product vouchers with pre-checkout margin check.	WF-03	Nguyễn Đình Tuấn

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-10	Multi-Store Order Splitting & In-place Quick Login	Customer	Atomic DB order splitting per store, in-place zero-loss Quick Login Modal, transaction rollback on failure.	WF-03	Nguyễn Đình Tuấn
F-11	6-Block Merchant Product Variant Matrix	Shop Manager, System Manager	Shopee-standard 6-block layout: basic info, media, 2-tier variant pricing matrix, logistics, commission, legal & moderation.	WF-04	Nguyễn Phú Quý
F-12	Merchant Order Lifecycle & GHN Fulfillment	Shop Manager, Customer	GHN Sandbox parcel creation, dynamic weight/dimension fee calculation, live webhook tracking.	WF-04	Nguyễn Phú Quý
F-13	14-Day Adaptive Escrow Commission Holding Ledger	Shop Manager, Collaborator, System Manager	Holds commission in escrow post-delivery for 14 days (matching customer return SLA); automated release and return clawback.	WF-04	Nguyễn Phú Quý
F-14	Return, Refund & Dual-Video Dispute Mediation Center	Customer, Shop Manager, System Manager	Customer return request flow, GHN reverse pickup, 48h shop inspection SLA, System Manager dual-video escrow arbitration.	WF-04	Nguyễn Phú Quý
F-15	Collaborator Sample Request & 14-Day Review SLA	Collaborator, Shop Manager, System Manager	Sample product request, quota throttling, delivery tracking, 14-day review content proof SLA with System Manager arbitration.	WF-02	Phan Xuân Thịnh
F-16	Real-time Chat & Exclusive Deal Negotiation	Collaborator, Shop Manager	Socket.IO real-time chat, rich proposal card, bi-directional exclusive commission negotiation.	WF-02	Phan Xuân Thịnh
F-17	Interactive Livestream Shopping Room	Collaborator, Customer	WebRTC/HLS room, dynamic live deal pinning, real-time live voucher drop, conversion metrics.	WF-02	Phan Xuân Thịnh
F-18	Realtime Analytics & Leaderboard with AI Fraud	Collaborator, Shop Manager, System Manager, System Administrator	Real-time GMV/commission dashboards, gamified tier ranking, AI traffic fraud sentinel.	WF-05	Nguyễn Đình Tuấn
F-19	Concurrency-Safe Wallet Withdrawal & Payout Management	Collaborator, Shop Manager, System Manager	Automated withdrawal request with DB row-level locking, 10% PIT tax withholding, VietQR batch export, and bank bill verification.	WF-05	Nguyễn Đình Tuấn & Nguyễn Thành Thắng

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-01	Performance & Latency	API Response Time (p95)	<= 200 ms for catalog browsing and tracking redirection	Autocannon load test (1,000 req/s) & NestJS duration interceptor	Nguyễn Thành Thắng
NFR-02	Security & Password Hashing	Argon2id Hash & JWT Lifespan	Argon2id (>= 64MB memory, 3 iterations); JWT access token 15 min	Automated auth test suites (auth-otp.spec.ts) & OWASP ZAP scan	Nguyễn Thành Thắng
NFR-03	Availability & Transaction Safety	Uptime & Atomic Rollback Rate	99.5% service uptime; 100% atomic rollback on stock exhaustion	Jest E2E order rollback simulation & /health endpoint monitoring	Nguyễn Đình Tuấn
NFR-04	Frontend Web Vitals	Bundle Size & First Contentful Paint	Initial bundle <= 500 KB (brotli); FCP <= 1.2s; CLS < 0.1	Vite build bundle visualizer & Google Lighthouse CI audit	Phan Xuân Thịnh
NFR-05	Concurrency & Rate Limiting	Sliding Window Limit & Failover	100 req/min/IP with zero crash on Redis outage (RAM fallback)	Jest unit safety tests (jest-unit-safety.spec.ts) & Apache Bench	Nguyễn Đình Tuấn
NFR-06	Financial Data Auditability	Audit Log Completeness	100% of financial, commission, and dispute movements logged	Prisma audit interceptor verification test (audit.interceptor.spec.ts)	Nguyễn Phú Quý

5. Out of Scope

Item	Reason
Proprietary Delivery Fleet & Physical Couriers	Delegated to Giao Hàng Nhanh (GHN) Sandbox API with automatic shipping fee calculation and parcel tracking.

Item	Reason
Direct Fiat Commercial Banking Gateway	Delegated to PayOS / VietQR payment gateway sandbox; interbank license is out of university capstone scope.
On-premise Video Transcoding CDN Infrastructure	Delegated to WebRTC peer streaming and simulated HLS chunk distribution for livestream shopping rooms.
Enterprise ERP / WMS Physical Warehouse System	Platform manages digital inventory, variant matrices, and logistics orders without proprietary robotics hardware.

6. Workflow Commitments

6.1 Workflow WF-01

Workflow name	WF-01: User Registration, 2FA Authentication & Multi-tier KYC Approval
Owner (one member)	Nguyễn Thành Thắng
Supporting members	Nguyễn Đình Tuấn, Phan Xuân Thịnh
Actors	Collaborator, Shop Manager, System Manager
Trigger	Collaborator or Shop Manager registers account, verifies OTP, and submits identity dossier
Business outcome	Verified user account with role-based portal access, KYC verified badge, full affiliate capabilities unlocked
Main entity state path	User: PENDING -> ACTIVE; KYC: PENDING_REVIEW -> VERIFIED; Tier: SILVER
Activity diagram (Report 4 section)	Report 4 Section 3.1 (Activity Diagram AD-WF01)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-01	User Authentication & RBAC Guard	Argon2id hashing, JWT Access/Refresh tokens, RBAC guard for 5 roles.	TC-F01-01..03
F-02	Identity Verification & KYC Multi-Tier Approval	CCCD / Business license upload, multi-status verification (PENDING/APPROVED/REJECTED) by System Manager.	TC-F02-01..03
F-03	Collaborator Social Portfolio Linking	Link TikTok, Facebook, YouTube channels with follower counts and content category tags.	TC-F03-01..03
F-04	Digital Media Asset Hub	Centralized promotional asset library with tags, resolution filtering, and one-click download.	TC-F04-01..03
F-05	Real-time Security Audit Logging	Immutable recording of sensitive financial, KYC, payout, and dispute actions with actor ID & IP.	TC-F05-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	**Main**	TC_KYC_01..04: Collaborator or Shop Manager registers with phone/email, verifies OTP, submits 12-digit ID with 2 photos.	1. Register account with Email & Phone. 2. Submit 6-digit OTP. 3. Upload CCCD & Bank details. 4. System Manager audits & approves.	Account status ACTIVE; KYC badge marked VERIFIED; role portal access unlocked (TC_KYC_01..04).	Nguyễn Thành Thắng
DS-01b	**Exception**	TC_KYC_01..04: Duplicate email / weak password / >5 wrong OTP locks 15m OR blurred ID photo.	1. Submit duplicate email. 2. Input wrong OTP 5 times. 3. System Manager clicks Reject with mandatory feedback note.	Account locked 15 mins; KYC status transitions to REJECTED; resubmission unlocked (TC_KYC_01..04).	Nguyễn Thành Thắng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	15	Yes (100%)	Nguyễn Thành Thắng
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Thành Thắng
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Thành Thắng
Total for this workflow	**19**	**Yes (100%)**	**Nguyễn Thành Thắng**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes
Main scenario passes	**Yes**
Exception scenarios passing	**100% (1/1)**
Test pass rate for this workflow	**100%**
Open Critical / High defects	**0**
Open Critical / High defects	*0*

Reviewer note

Reviewer note	Fully demonstrated end-to-end with automated test coverage in CI.

6.2 Workflow WF-02

Workflow name	WF-02: Exclusive Deal Negotiation, In-App Chat & Live Commerce Deal Pinning
Owner (one member)	Phan Xuân Thịnh
Supporting members	Nguyễn Thành Thắng, Nguyễn Phú Quý
Actors	Collaborator, Shop Manager, Customer
Trigger	Shop Manager publishes 10% base product; Collaborator proposes 13% VIP rate via Chat OR Shop Manager invites Collaborator; Collaborator starts live session
Business outcome	VIP Deal agreed, VIP Voucher Code & Smart Link issued, live session with deal pinning (+3% live bonus, e.g. 16%)
Main entity state path	Deal: PROPOSED -> AGREED; LiveSession: SCHEDULED -> LIVE -> ENDED; Pin: PINNED -> UNPINNED
Activity diagram (Report 4 section)	Report 4 Section 3.2 (Activity Diagram AD-WF02)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-06	Additive Multi-Rule Commission Engine	Calculates min(25%, Base(10%) + Tier(1-3%) + max(Deal, Campaign) + Live(2-3%)) + MonthlyBonus.	TC-F06-01..03
F-15	Collaborator Sample Request & 14-Day Review SLA	Sample product request, quota throttling, delivery tracking, 14-day review content proof SLA with System Manager arbitration.	TC-F15-01..03
F-16	Real-time Chat & Exclusive Deal Negotiation	Socket.IO real-time chat, rich proposal card, bi-directional exclusive commission negotiation.	TC-F16-01..03
F-17	Interactive Livestream Shopping Room	WebRTC/HLS room, dynamic live deal pinning, real-time live voucher drop, conversion metrics.	TC-F17-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02a	**Main**	TC_DEAL_01..04: Shop Manager proposes 13% VIP rate via Chat; Collaborator clicks Accept; Collaborator goes LIVE and pins deal.	1. Shop Manager selects product, proposes 13% rate in Chat. 2. Collaborator clicks Accept Deal. 3. Collaborator goes LIVE. 4. Pins deal (+3% live bonus).	VIP voucher generated; live stream active with +3% live bonus (16% total); summary report on stream end (TC_DEAL_01..04).	Phan Xuân Thịnh
DS-02b	**Exception**	TC_DEAL_01..04: Proposal rate > 25% ceiling blocked; Collaborator rejects deal (retains base 10%); live item out of stock.	1. Input rate > 25%. 2. Collaborator clicks Reject. 3. Product stock depleted to 0 during live broadcast.	Submission blocked by 25% ceiling cap; deal canceled retaining base 10%; purchase button hidden in live tray (TC_DEAL_01..04).	Phan Xuân Thịnh

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	12	Yes (100%)	Phan Xuân Thịnh
End-to-end workflow test (>= 1)	1	Yes (100%)	Phan Xuân Thịnh

Test type	Count	Automated in CI	Responsible member(s)
State transition tests (>= 1 per transition)	3	Yes (100%)	Phan Xuân Thịnh
Total for this workflow	**16**	**Yes (100%)**	**Phan Xuân Thịnh**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes
Main scenario passes	**Yes**
Exception scenarios passing	**100% (1/1)**
Test pass rate for this workflow	**100%**
Open Critical / High defects	**0**
Open Critical / High defects	*0*

Reviewer note

Reviewer note	Fully demonstrated end-to-end with automated test coverage in CI.

6.3 Workflow WF-03

Workflow name	WF-03: Multi-Channel Attribution, Quick Login Modal & VietQR Checkout
Owner (one member)	Nguyễn Đình Tuấn
Supporting members	Nguyễn Thành Thắng, Nguyễn Phú Quý
Actors	Customer, Collaborator, Shop Manager, Payment Gateway
Trigger	Customer clicks Collaborator Smart Link / QR / Live Deal, adds multi-store items, and clicks Checkout
Business outcome	30-day marketing cookie recorded, 100% cart preserved during in-place login modal, instant VietQR payment verification, order PAID
Main entity state path	Link: ACTIVE; Cookie: SET (30d); Order: PENDING_PAYMENT -> PROCESSING (PAID); Cart: PRESERVED
Activity diagram (Report 4 section)	Report 4 Section 3.3 (Activity Diagram AD-WF03)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-06	Additive Multi-Rule Commission Engine	Calculates min(25%, Base(10%) + Tier(1-3%) + max(Deal, Campaign) + Live(2-3%)) + MonthlyBonus.	TC-F06-01..03
F-07	Smart Shortened Link & QR Generator	Generates /r/:code links and SVG dynamic QR codes with Redis rate limit protection.	TC-F07-01..03
F-08	Last-Click Cookie Attribution & Tracking	30-day cookie window, first/last touch attribution, referrer preservation across sessions.	TC-F08-01..03
F-09	3-Tier Multi-Store Voucher Stacking	Stack Global + Store (couponsByStore) + Product vouchers with pre-checkout margin check.	TC-F09-01..03
F-10	Multi-Store Order Splitting & In-place Quick Login	Atomic DB order splitting per store, in-place zero-loss Quick Login Modal, transaction rollback on failure.	TC-F10-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03a	**Main**	TC_ATTR_01..04: Collaborator generates /r/:slug smart link; Customer clicks (cookie set 30 days); adds multi-store items; checks out.	1. Collaborator generates smart link & QR. 2. Customer clicks link. 3. Adds items from Shop A & B. 4. Quick login in-place. 5. Scans VietQR.	30-day marketing cookie recorded; 100% cart preserved during login modal; PayOS webhook marks order PAID (TC_ATTR_01..04).	Nguyễn Đình Tuấn
DS-03b	**Exception**	TC_ATTR_01..04: Unverified KYC blocks link creation; Customer visits deleted item (404); VietQR underpaid.	1. Unverified Collaborator attempts link creation. 2. Customer visits deleted SKU. 3. Customer underpays VietQR invoice.	Link button disabled with KYC prompt; 404 Discontinued page displayed; underpaid order remains in Pending Payment (TC_ATTR_01..04).	Nguyễn Đình Tuấn

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	15	Yes (100%)	Nguyễn Đình Tuấn
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Đình Tuấn
State transition tests (>= 1 per transition)	2	Yes (100%)	Nguyễn Đình Tuấn
Total for this workflow	**18**	**Yes (100%)**	**Nguyễn Đình Tuấn**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes
Main scenario passes	**Yes**
Exception scenarios passing	**100% (1/1)**
Test pass rate for this workflow	**100%**
Open Critical / High defects	**0**
Open Critical / High defects	*0*

Reviewer note

Reviewer note	Fully demonstrated end-to-end with automated test coverage in CI.

6.4 Workflow WF-04

Workflow name	WF-04: GHN Logistics Fulfillment, 14-Day Adaptive Escrow & Dual-Video Return Arbitration
Owner (one member)	Nguyễn Phú Quý
Supporting members	Nguyễn Đình Tuấn, Phan Xuân Thịnh
Actors	Shop Manager, Customer, GHN Logistics, System Manager
Trigger	Shop Manager confirms order, creates GHN shipment parcel, delivers to Customer; Customer requests return within 14 days
Business outcome	GHN waybill generated, delivery completed, 14-day adaptive escrow quarantine active; if return disputed: dual-video arbitration, 100% Customer wallet refund
Main entity state path	Order: PROCESSING -> SHIPPED -> DELIVERED -> COMPLETED; Escrow: LOCKED -> RELEASED/CLAWED_BACK
Activity diagram (Report 4 section)	Report 4 Section 3.4 (Activity Diagram AD-WF04)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-11	6-Block Merchant Product Variant Matrix	Shopee-standard 6-block layout: basic info, media, 2-tier variant pricing matrix, logistics, commission, legal & moderation.	TC-F11-01..03
F-12	Merchant Order Lifecycle & GHN Fulfillment	GHN Sandbox parcel creation, dynamic weight/dimension fee calculation, live webhook tracking.	TC-F12-01..03
F-13	14-Day Adaptive Escrow Commission Holding Ledger	Holds commission in escrow post-delivery for 14 days (matching customer return SLA); automated release and return clawback.	TC-F13-01..03
F-14	Return, Refund & Dual-Video Dispute Mediation Center	Customer return request flow, GHN reverse pickup, 48h shop inspection SLA, System Manager dual-video escrow arbitration.	TC-F14-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04a	**Main**	TC_FUL_01..04: Shop Manager packages order with valid weight/dimensions, creates GHN parcel, Customer receives order.	1. Shop Manager confirms order. 2. Confirms weight/dimensions. 3. Calls GHN Sandbox API. 4. Delivery completed.	GHN tracking code generated (e.g. GHN-TEST-12345); status In Transit -> Delivered; 14-day adaptive escrow quarantine active (TC_FUL_01..04).	Nguyễn Phú Quý
DS-04b	**Exception**	TC_FUL_01..04: Customer submits return within 14 days with unboxing video; Shop Manager rejects dispute; System Manager arbitrates.	1. Customer requests return with video. 2. Shop Manager rejects dispute. 3. System Manager reviews unboxing vs packaging videos.	Escrow frozen; System Manager approves Customer claim: 100% wallet refund, commission clawed back (TC_FUL_01..04).	Nguyễn Phú Quý

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	12	Yes (100%)	Nguyễn Phú Quý
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Phú Quý
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Phú Quý
Total for this workflow	**16**	**Yes (100%)**	**Nguyễn Phú Quý**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes
Main scenario passes	**Yes**
Exception scenarios passing	**100% (1/1)**
Test pass rate for this workflow	**100%**
Open Critical / High defects	**0**
Open Critical / High defects	*0*

Reviewer note

Reviewer note	Fully demonstrated end-to-end with automated test coverage in CI.

6.5 Workflow WF-05

Workflow name	WF-05: Multi-Role Wallet Withdrawal, 10% PIT Tax Withholding & Concurrency Payout Settlement
Owner (one member)	Nguyễn Đình Tuấn & Nguyễn Thành Thắng
Supporting members	Nguyễn Phú Quý, Phan Xuân Thịnh
Actors	Collaborator, Shop Manager, System Manager, Banking Gateway
Trigger	Collaborator or Shop Manager submits withdrawal request, or month-end bonus scan calculates GMV milestone
Business outcome	10% statutory PIT tax withheld, available balance locked via database row-level locking, VietQR batch exported, bank bill verified, and funds disbursed via Napas 24/7
Main entity state path	Wallet: LOCKED -> DEBITED; Withdrawal: PENDING -> PROCESSING -> APPROVED -> DISBURSED; Tax: WITHHELD
Activity diagram (Report 4 section)	Report 4 Section 3.5 (Activity Diagram AD-WF05)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-05	Real-time Security Audit Logging	Immutable recording of sensitive financial, KYC, payout, and dispute actions with actor ID & IP.	TC-F05-01..03
F-06	Additive Multi-Rule Commission Engine	Calculates min(25%, Base(10%) + Tier(1-3%) + max(Deal, Campaign) + Live(2-3%)) + MonthlyBonus.	TC-F06-01..03

Feature ID	Feature	Acceptance criteria	Test case IDs
F-18	Realtime Analytics & Leaderboard with AI Fraud	Real-time GMV/commission dashboards, gamified tier ranking, AI traffic fraud sentinel.	TC-F18-01..03
F-19	Concurrency-Safe Wallet Withdrawal & Payout Management	Automated withdrawal request with DB row-level locking, 10% PIT tax withholding, VietQR batch export, and bank bill verification.	TC-F19-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05a	**Main**	TC_PAY_01..04: Collaborator or Shop Manager requests payout; 10% PIT tax withheld; balance locked via DB row-level lock; System Manager audits recipient bank account, uploads bank transfer bill, and dispatches via Napas 24/7.	1. Collaborator or Shop Manager submits withdrawal request. 2. 10% PIT tax withheld and available balance locked. 3. System Manager reviews bank account details and attaches bank transfer receipt. 4. Dispatches payout via Napas 24/7.	Net amount disbursed to bank; 10% PIT tax withheld; transaction receipt attached; locked balance permanently debited (TC_PAY_01..04).	Nguyễn Đình Tuấn & Nguyễn Thành Thắng
DS-05b	**Exception**	TC_PAY_01..04: Insufficient balance (< 100k or > balance); unauthorized access; invalid bank account; bank gateway failure.	1. Request amount > available balance. 2. Unauthorized role attempts approval. 3. Bank gateway returns transfer failure.	Submission blocked; 403 Forbidden on unauthorized audit; 100% requested funds safely restored to wallet with error logged (TC_PAY_01..04).	Nguyễn Đình Tuấn & Nguyễn Thành Thắng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	12	Yes (100%)	Nguyễn Đình Tuấn & Nguyễn Thành Thắng
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Đình Tuấn & Nguyễn Thành Thắng
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Đình Tuấn & Nguyễn Thành Thắng
Total for this workflow	**16**	**Yes (100%)**	**Nguyễn Đình Tuấn & Nguyễn Thành Thắng**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes
Main scenario passes	**Yes**
Exception scenarios passing	**100% (1/1)**
Test pass rate for this workflow	**100%**
Open Critical / High defects	**0**
Open Critical / High defects	*0*

Reviewer note

Reviewer note	Fully demonstrated end-to-end with automated test coverage in CI.

7. Individual Commitments

One block per member. Every member owns at least one main workflow and 2–3 parts. Each test case and each fix is counted for one member only.

7.1 Member 1

Full name	**Nguyễn Thành Thắng**
Student ID	SE184251
Role in team	Team Leader & Fullstack Architect

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-01: User Registration, 2FA Authentication & Multi-tier KYC Approval; WF-05: Multi-Role Wallet Withdrawal & Payouts (Co-owner)	Primary owner
Mandatory features owned	F-01 (Auth & RBAC), F-02 (KYC Multi-Tier), F-03 (Social Portfolio), F-04 (Media Hub), F-05 (Audit Logging)	Core responsibility
Modules / entities owned	AuthModule, KycModule, MediaModule, AuditModule, UserModule	Code implementation
Design decisions owned	D2: Strict Authenticated Checkout with Zero-Loss Cart; PR #1, PR #5	Architectural design
NFRs responsible	NFR-01 (API Latency <= 200ms), NFR-02 (Argon2id Hash & JWT Lifespan)	Quality assurance

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	22 functional test cases (100% automated in CI)	Review 2 (Done)
Of which automated in CI	22 (100%)	GitHub Actions CI
Demo scenarios presented	DS-01a, DS-01b, DS-05a (TC_KYC_01..04, TC_PAY_01..04)	Review 2 / Defense 1.1
Required fixes owned	All assigned defects closed (0 open)	Review 2 (Done)
Documentation sections	Report 4 / SDD sections for Team Leader & Fullstack Architect	Review 2 (Done)

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☑ Verified ☐ Partly verified ☐ Not verified *☐ Partly verified* *☐ Not verified*	None. Code clean, tests pass in CI, demo scripts verified.	Ready for Defense 1.1.

Member confirmation

Signature / date	Confirmed by Nguyễn Thành Thắng · 29/10/2026

7.2 Member 2

Full name	**Phan Xuân Thịnh**
Student ID	SE184527
Role in team	Fullstack Developer & Realtime Lead

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-02: Exclusive Deal Negotiation, In-App Chat & Live Commerce Deal Pinning	Primary owner
Mandatory features owned	F-06 (Commission Engine), F-15 (Sample Request & 14-Day Review SLA), F-16 (Deal Chat Socket.IO), F-17 (Livestream Shopping Room)	Core responsibility
Modules / entities owned	SamplesModule, ChatModule, LiveSessionsModule, CommissionRulesModule	Code implementation
Design decisions owned	D5: Dual-Channel Real-time Live & Messaging Gateway; PR #16, PR #27	Architectural design
NFRs responsible	NFR-04 (Frontend Bundle <= 500KB, FCP <= 1.2s)	Quality assurance

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	18 functional test cases (100% automated in CI)	Review 2 (Done)
Of which automated in CI	18 (100%)	GitHub Actions CI
Demo scenarios presented	DS-02a, DS-02b (TC_DEAL_01..04)	Review 2 / Defense 1.1
Required fixes owned	All assigned defects closed (0 open)	Review 2 (Done)
Documentation sections	Report 4 / SDD sections for Fullstack Developer & Realtime Lead	Review 2 (Done)

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☑ Verified ☐ Partly verified ☐ Not verified *☐ Partly verified* *☐ Not verified*	None. Code clean, tests pass in CI, demo scripts verified.	Ready for Defense 1.1.

Member confirmation

Signature / date	Confirmed by Phan Xuân Thịnh · 29/10/2026

7.3 Member 3

Full name	**Nguyễn Đình Tuấn**
Student ID	SE182540
Role in team	Backend Lead & Financial Engine Specialist

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-03: Multi-Channel Attribution, Quick Login Modal & VietQR Checkout; WF-05: Multi-Role Wallet Withdrawal & Payouts (Co-owner)	Primary owner
Mandatory features owned	F-07 (Smart Link & QR), F-08 (Last-Click Attribution), F-09 (Voucher Stacking), F-10 (Order Splitting & Checkout), F-18 (Analytics & AI Fraud), F-19 (Concurrency-Safe Payouts)	Core responsibility
Modules / entities owned	ReferralLinksModule, CouponsModule, CheckoutModule, DashboardModule, AiModule, PayoutsModule	Code implementation
Design decisions owned	D1: Additive Commission Engine; D4: 3-Tier Multi-Store Voucher Hierarchy; PR #9, PR #12	Architectural design
NFRs responsible	NFR-03 (99.5% Uptime & Atomic Rollback), NFR-05 (100 req/min Rate Limiting)	Quality assurance

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	23 functional test cases (100% automated in CI)	Review 2 (Done)
Of which automated in CI	23 (100%)	GitHub Actions CI
Demo scenarios presented	DS-03a, DS-03b, DS-05a, DS-05b (TC_ATTR_01..04, TC_PAY_01..04)	Review 2 / Defense 1.1
Required fixes owned	All assigned defects closed (0 open)	Review 2 (Done)
Documentation sections	Report 4 / SDD sections for Backend Lead & Financial Engine Specialist	Review 2 (Done)

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☑ Verified ☐ Partly verified ☐ Not verified *☐ Partly verified* *☐ Not verified*	None. Code clean, tests pass in CI, demo scripts verified.	Ready for Defense 1.1.

Member confirmation

Signature / date	Confirmed by Nguyễn Đình Tuấn · 29/10/2026

7.4 Member 4

Full name	**Nguyễn Phú Quý**
Student ID	SE180104
Role in team	Fullstack Developer & Logistics Lead

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-04: GHN Logistics Fulfillment, 14-Day Adaptive Escrow & Dual-Video Return Arbitration	Primary owner
Mandatory features owned	F-11 (Product Variant Matrix), F-12 (Order Lifecycle & GHN), F-13 (14-Day Adaptive Escrow Ledger), F-14 (Dual-Video Return Mediation)	Core responsibility
Modules / entities owned	ProductsModule, OrdersModule, ShippingModule, ReturnsModule, WalletsModule	Code implementation
Design decisions owned	D3: 14-Day Adaptive Escrow Commission Quarantine & Dual-Video Dispute Arbitration; PR #11, PR #14	Architectural design
NFRs responsible	NFR-06 (100% Financial Ledger Auditability)	Quality assurance

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	18 functional test cases (100% automated in CI)	Review 2 (Done)
Of which automated in CI	18 (100%)	GitHub Actions CI
Demo scenarios presented	DS-04a, DS-04b (TC_FUL_01..04)	Review 2 / Defense 1.1
Required fixes owned	All assigned defects closed (0 open)	Review 2 (Done)
Documentation sections	Report 4 / SDD sections for Fullstack Developer & Logistics Lead	Review 2 (Done)

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☑ Verified ☐ Partly verified ☐ Not verified *☐ Partly verified* *☐ Not verified*	None. Code clean, tests pass in CI, demo scripts verified.	Ready for Defense 1.1.

Member confirmation

Signature / date	Confirmed by Nguyễn Phú Quý · 29/10/2026

7.5 Member 5

Full name	N/A (Finalized 4-member project team)
Student ID	N/A
Role in team	N/A

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned		
Mandatory features owned		
Modules / entities owned		
Design decisions owned (D-IDs)		
NFRs responsible		

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written		
Of which automated in CI		
Demo scenarios presented		
Required fixes owned (section 9)		
Documentation sections (Report / section)		

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01	**Nguyễn Thành Thắng**	F-01, F-02, F-03, F-04, F-05	DS-01a (Main), DS-01b (Exception)	19	Yes (100%)
WF-02	**Phan Xuân Thịnh**	F-06, F-15, F-16, F-17	DS-02a (Main), DS-02b (Exception)	16	Yes (100%)
WF-03	**Nguyễn Đình Tuấn**	F-06, F-07, F-08, F-09, F-10	DS-03a (Main), DS-03b (Exception)	18	Yes (100%)
WF-04	**Nguyễn Phú Quý**	F-11, F-12, F-13, F-14	DS-04a (Main), DS-04b (Exception)	16	Yes (100%)
WF-05	**Nguyễn Đình Tuấn & Nguyễn Thành Thắng**	F-05, F-06, F-18, F-19	DS-05a (Main), DS-05b (Exception)	16	Yes (100%)

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Nguyễn Thành Thắng (Leader)	WF-01, WF-05	F-01, F-02, F-03, F-04, F-05	22	Yes (100%)	DS-01a, DS-01b, DS-05a	Verified
Phan Xuân Thịnh	WF-02	F-06, F-15, F-16, F-17	18	Yes (100%)	DS-02a, DS-02b	Verified
Nguyễn Đình Tuấn	WF-03, WF-05	F-07, F-08, F-09, F-10, F-18, F-19	23	Yes (100%)	DS-03a, DS-03b, DS-05a, DS-05b	Verified
Nguyễn Phú Quý	WF-04	F-11, F-12, F-13, F-14	18	Yes (100%)	DS-04a, DS-04b	Verified

The totals in 8.1 and 8.2 must both equal the committed numbers in 8.3.

8.3 Committed Numbers

Number of main workflows	**5**
Number of demo scenarios (main / exception)	**10 (5 main / 5 exception)**
Committed functional test cases	**81 (1,514 automated tests in CI)**
Percentage automated in CI	**> 95% (Minimum requirement >= 50%)**

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	100% run end-to-end	100% verified on staging with seeded master data
Demo scenarios	100% main; >= 80% exception	100% main (5/5); 100% exception (5/5) verified
Committed test cases pass rate	>= 90% at Defense 1.1	100% currently passing in CI (1,514/1,514 passed)
Automated in CI	>= 50% of committed total	> 95% automated in GitHub Actions CI
Known Gaps closed or planned	All High findings addressed	100% High findings from BM02 closed (7/7 Fixed)

Criterion	Minimum threshold	Agreed threshold
Code quality & build integrity	Zero compiler errors	0 TypeScript errors (tsc --noEmit); 0 Vite build errors
Brand & Design tokens	Warm Sand & Brand Gold	100% compliant (#FAF8F5, #F3EFE6, #C59B58, #B88E4F)

9. Required Fixes

Items from the Review 2 minutes and Planned items from Form BM02.

ID	Item	Related workflow	Owner	Due
G1	Automated Fiat Interbank Payout Disbursal vs Sandbox Simulation	WF-05	Nguyễn Phú Quý	Review 3
G2	In-browser WebRTC Peer Streaming vs Commercial CDN Media Cluster	WF-02	Phan Xuân Thịnh	Council 1.1
G3	Distributed Redis SPOF Graceful Degradation (LRU in-memory fallback)	WF-03	Nguyễn Đình Tuấn	Done

10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1	**longt5**		29/10/2026
Reviewer 2	**ThinhDP2**		29/10/2026
Supervisor	**ThS. Tôn Thất Hoàng Minh**		29/10/2026
Team leader	**Nguyễn Thành Thắng**		29/10/2026